

Sri Lanka Institute of Information Technology



Machine Learning Based Integrated Solution for Smart Pepper Farming in Sri Lanka

25-26J-479

Checklist 01

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GitHub Repository Link

<https://github.com/madaramegama7/Research-Project>

Sketches



Login

Email address

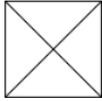
Password

☐

Login

Don't have an account? [Sign up](#)

[Forgot password?](#)



Sign Up

First Name

Last Name

Email address

Contact Number

Password

☐

Re-enter password

☐

Sign Up

Already have an account? [Log in](#)

Hello, Farmer
Ceylon Pepper

Mountain View
Tap to refresh
12°

5
Total Crops

2
Active Alerts

87.6%
Avg Quality

Smart Farming Tools

Yield Prediction
Forecast the harvest

Disease Detection
AI Diagnosis

Quality Grading
ISO standards

Market Forecast
Price Trends

More Services

AI Assistant
Explore

Marketplace
Explore

Home

Market

Insig

Settin

Hello, Exporter
Ceylon Pepper

Mountain View
Top Sri Lankan
12°C

12
Total Batches

2
Pending Orders

\$85 K
Total Worth

Export Services

Export Prices
Last 2 week

Quality Requests
3 Indicators

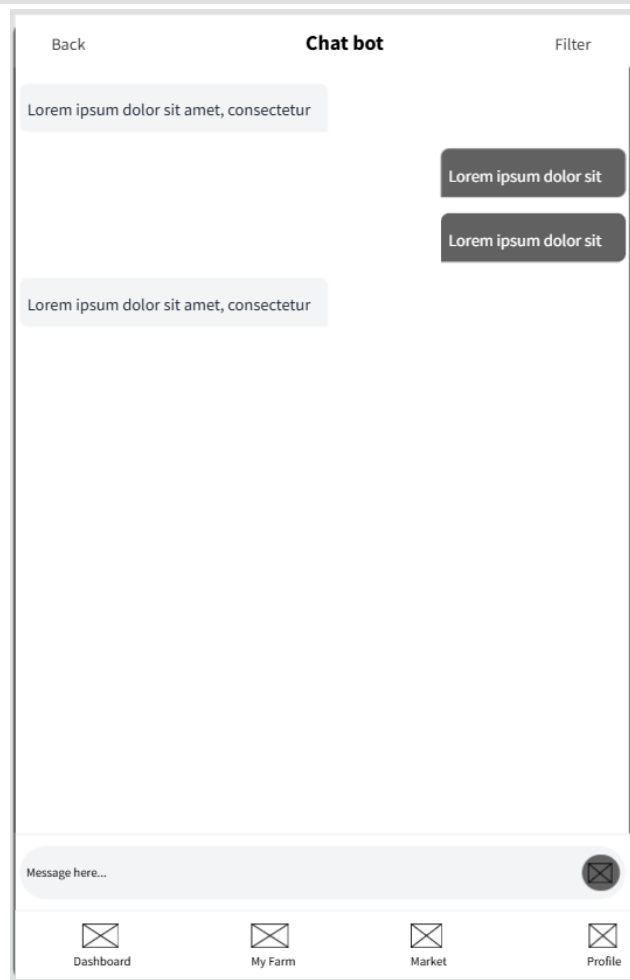
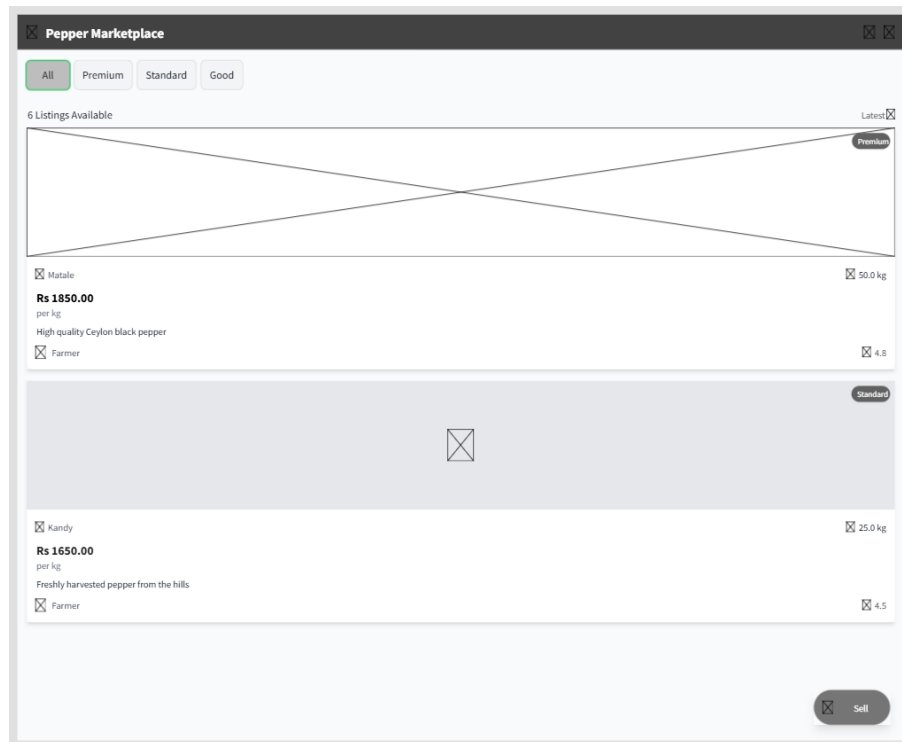
Export Batches
2 Shipments

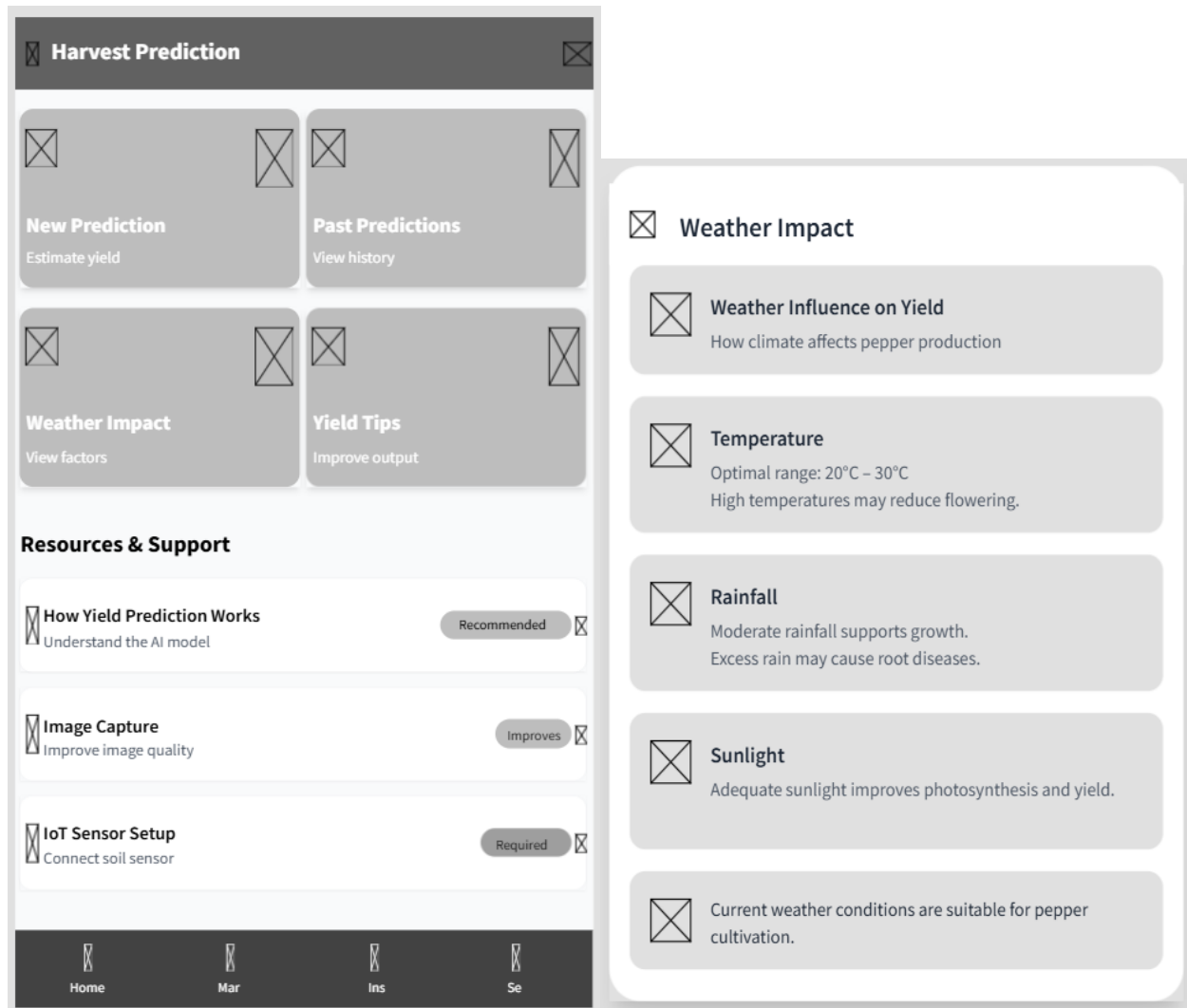
Traceability
Blockchain

Export Best Practices

Check global market demands

Maintain strict quality standards





✕

Market Forecast

Market Forecast

✕

Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut. Dolore

Explore Market Data

✕

Weekly Local Price Forecast

View weekly price predictions

✕

✕

Past Export Price Trends

Analyze export market trends

✕

✕

Export Details by Country

Track global exports

✕

✕

Dashboard

✕

My Farm

✕

Market

✕

Profile

✕

Weekly Price Forecast

District

Pepper Type

Grade

Week Details

Fetch Details

Weather Conditions

Fetch Details

Predict the Price

Price Forecast Analysis

January · Week 1, 2026

Predicted Price

Rs. 2150

+ Rs.260

Current vs Previous Price

Previous Week

Rs. 1800/kg

Current Week

Rs. 1890/kg

Past Price Trend

Last 5 weeks

● Black ● White

30

25

20

15

10

Ja

Ja

Ja

Fe

Fe

Market Insights

Price trend shows upward

Historical average: Rs. 1534/kg

Price range: Rs.1500 - Rs.2000

Recommendations

Back

Plant Analyse

19°

H:24° L:18°

Montreal, Canada

Mid Rain

*Attach maximum 5 photos

Drop here to attach or upload

Max size: 5GB

Analyse plant

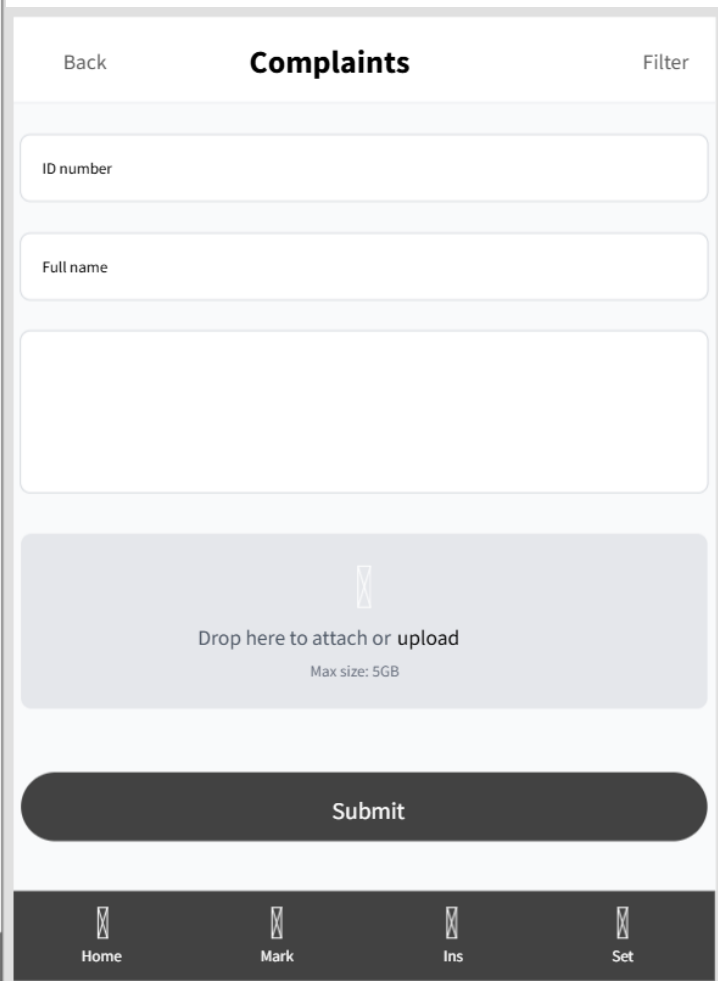
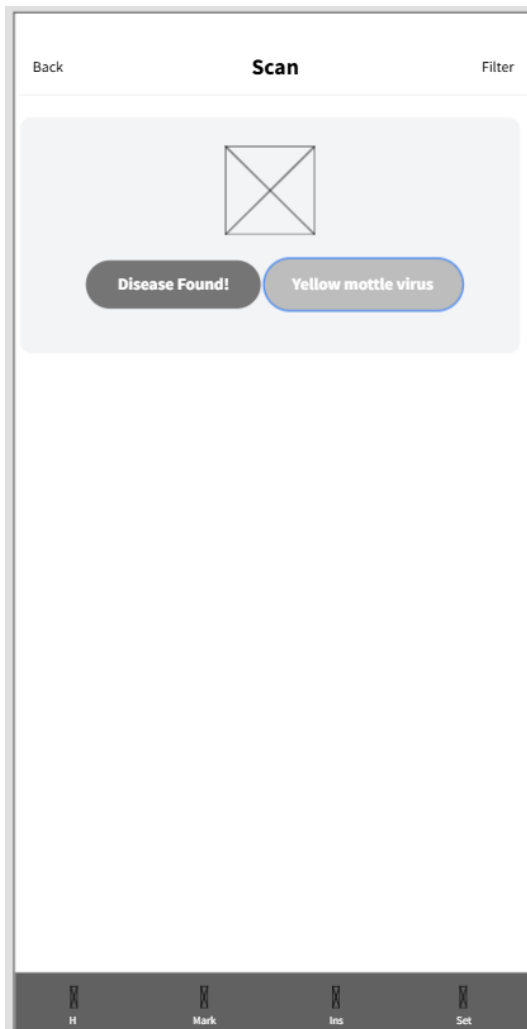
H

Mar

In

Se

9 | Page



UI Mockups



Create Your **Account**

First Name

 Enter your first name

Last Name

 Enter your last name

Email Address

 youremail@gmail.com

Contact Number

 0712345678

Password

 Create a strong password

Confirm Password



Welcome Back **Login**

Email Address

 youremail@gmail.com

Password

 Enter your password



☐ Remember me

[Forgot password?](#)

Login →

OR

 Continue with Google

Don't have an account? [Sign Up](#)

Hello, Farmer

Ceylon Pepper

Mountain View

Tap to refresh

12°C

5
Total Crops

2
Active Alerts

87.6%
Avg Quality

Smart Farming Tools

Yield Prediction
Forecast the harvest
→

Disease Detection
AI Diagnosis
→

Quality Grading
ISO standards
→

Market Forecast
Price Trends
→

More Services

AI Assistant
Explore →

Marketplace
Explore →

Farming Tips & Insights

Monitor soil moisture regularly

Apply organic fertilize monthly

Dashboard

My Farm

Market

Profile

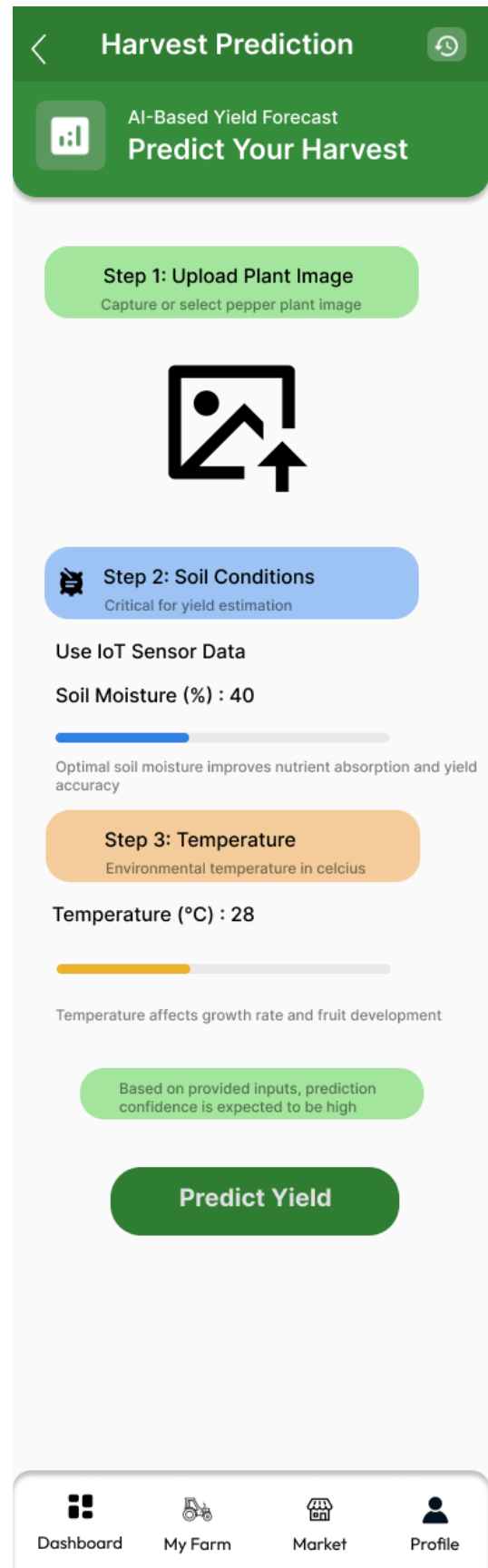
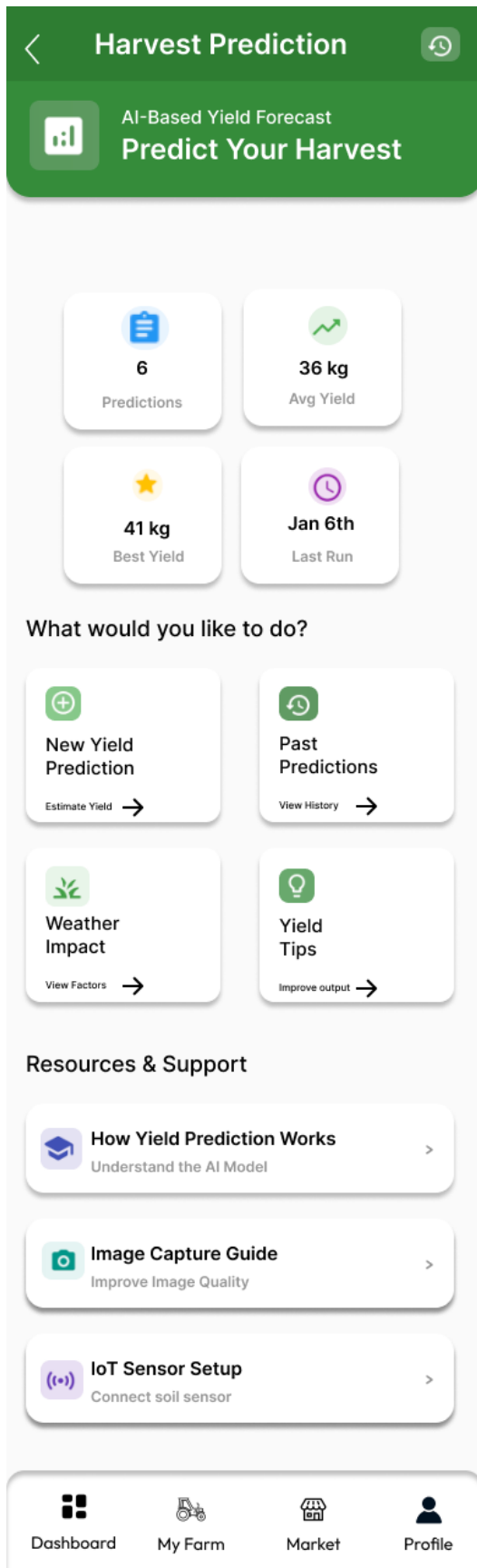
AI Farming Assistant

Always here to help

Hello! I'm your AI farming assistant. How can I help you with your black pepper cultivation today?

00:05

Ask me anything...



[Back](#)

Complaints

[Filter](#)

Drop here to attach or [upload](#)

Max size: 5GB

Submit Complaint



Home



Market Place



Insights



Settings

[Back](#)

Chat bot

[Filter](#)

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Donec fringilla quam eu faci
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Home



Market Place



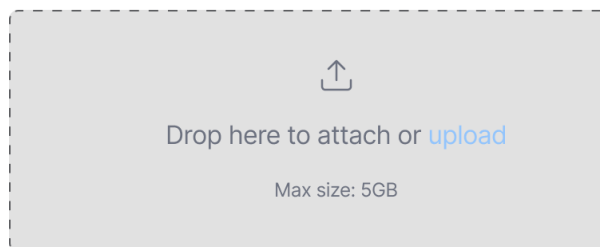
Insights



Settings

[Back](#)

Plant Analyse



Analyse plant



[Back](#)

Analyse Report

[Filter](#)

Disease percentage

87%

yellow mottle virus !

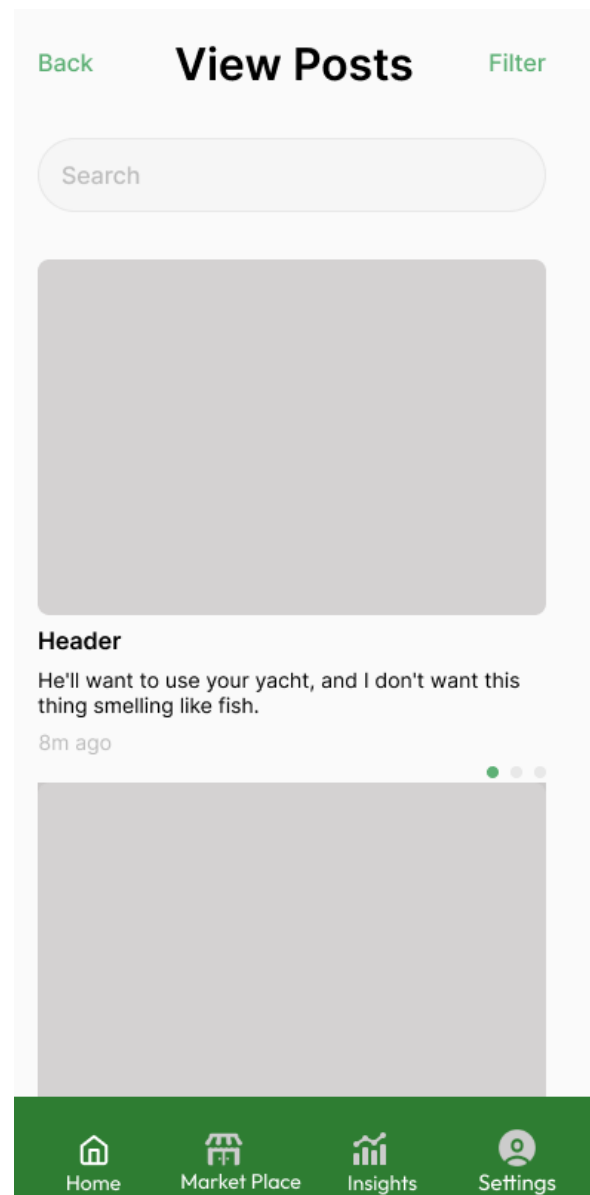
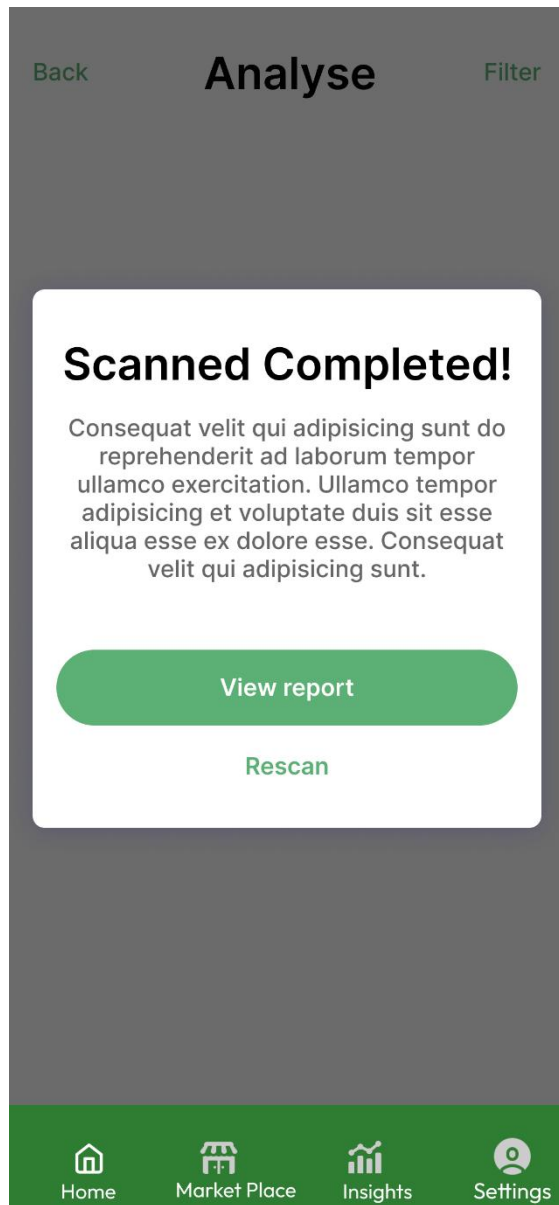


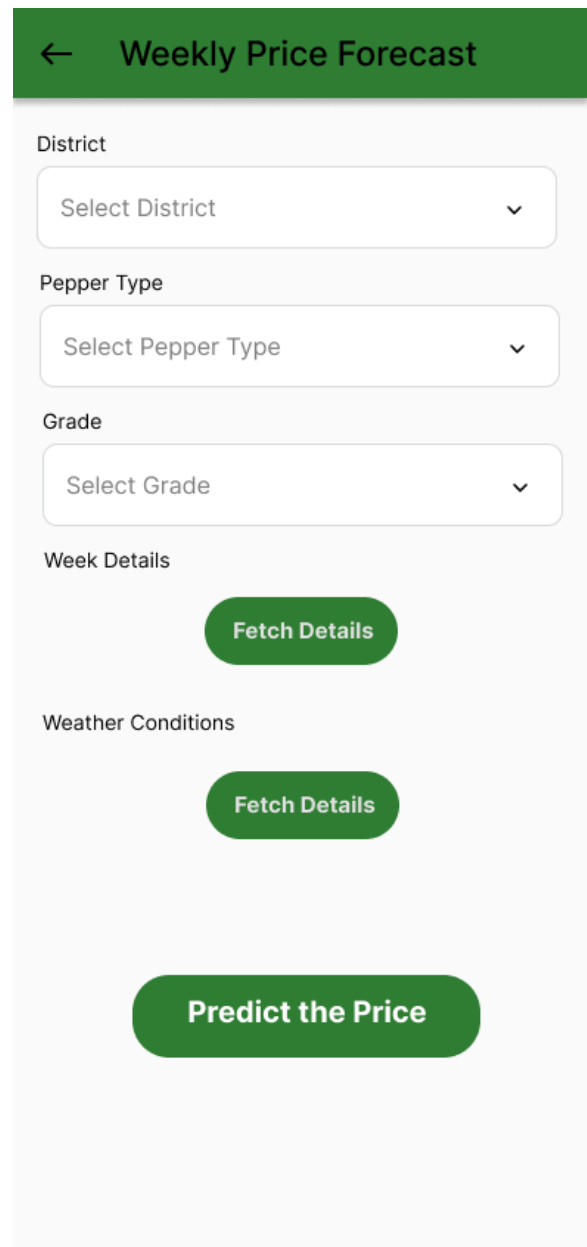
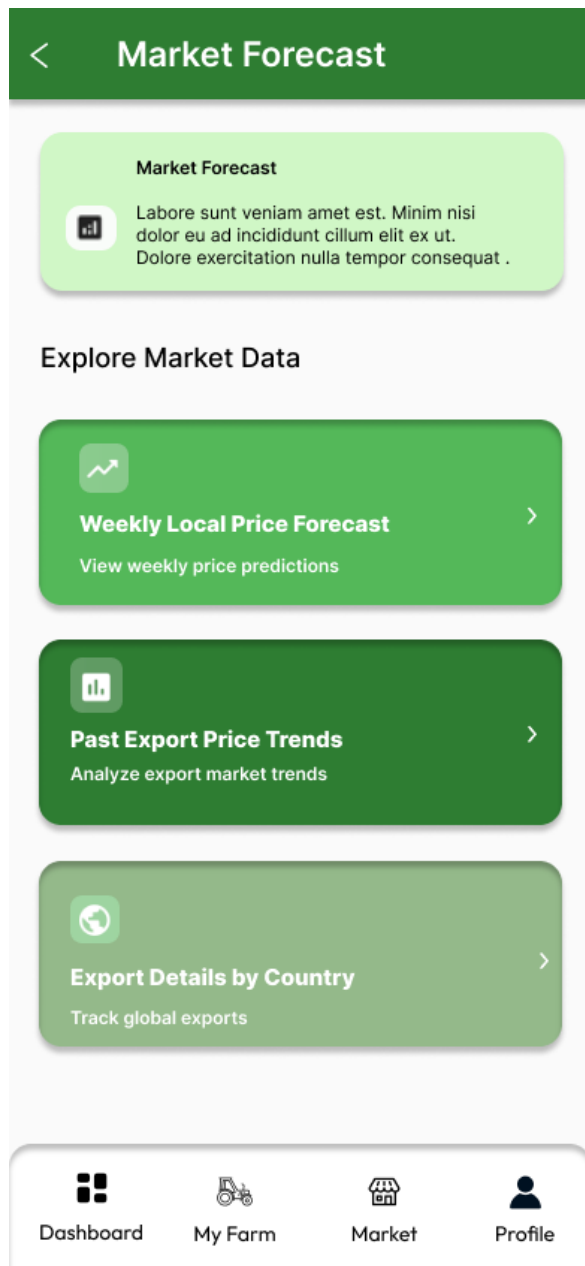
Contact
Officials



Download
report







← Price Forecast Analysis

January - Week 1, 2026

Predicted Price

Rs. **2150** /kg

Price Change
+ Rs.260 (13.8%)

Current vs Previous Price



Previous Week
Rs. 1800 /kg

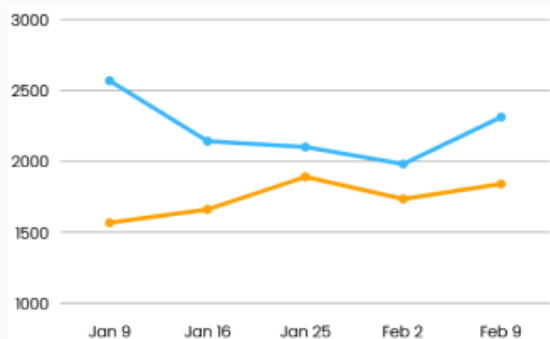


Current Week
Rs. 1890 /kg

Past Price Trend

Last 5 weeks comparison

● Black Pepper ● White Pepper



Market Insights

Price trend shows upward movement
Historical average: Rs. 1534/kg
Price range: Rs.1500 - Rs.2000

Recommendations

← Recommendations

Recommended Action

WAIT

Next week price is Rs.260 higher than current.
Wait to sell next week at Rs. 2150/kg.



Period

Jan 12 - Jan 18, 2026



Price Change

+13.8%

🕒 **Sell When Price Reaches**

Rs.2150/kg (next week forecast)

Monitor the market regularly for better selling opportunities

Price Analysis



Previous Week

Rs.1800/kg



Current Week

Rs.1890/kg



Next Week (Predicted)

Rs.2150/kg



Average Price

Rs.1730/kg

Market Insights

The market shows a strong upward trend with prices expected to increase significantly. This is a favorable time to hold your inventory for better returns.

✓ Recommended Actions

- 1 Store your pepper in a dry, cool place to maintain quality
- 2 Monitor market prices daily for optimal selling timing
- 3 Prepare logistics and transport arrangements in advance
- 4 Contact potential buyers to lock in selling arrangements

Past Export Price Trends

Historical Export Price Analysis



Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut. Dolore exercitation nulla tempor consequat.

Export Details By Country



Labore sunt veniam amet est. Minim nisi dolor eu ad.

[View →](#)

Filter Data

Year

From

To

Global Black Pepper Price Trends

Jan - Jul

Rs.3000

Rs.2500

Rs.2000

Rs.1500

Rs.1000

Jan Feb Mar Apr May Jun Jul

Peak Price

Rs. 2100/kg

Lowest Price

Rs. 1400/kg

Average

Rs. 1650/kg

Key Insights

Prices are trending upward

Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut.

Peak vs Average

Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut.

Momentum & Stability

Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut.

Recommendation

Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut.

Export Price Prediction



Export Price Prediction



Forecast export pepper prices monthly with your current details.

Fast 2-step setup

Realistic outlook

Get accurate monthly export price forecast to plan your shipments better

Pepper Type

Black

Export Volume (kg)

110

Year & Month

Predicting Price For

February 2026

Get export price predictions for this month

Predict the Export Price



Monthly Export Price Forecast

Your predicted export price

Pepper Type

Black

Month

February

Year

2026

Volume

110 kg

Per kg Price

Monthly Total

LKR

LKR

1480

177,860

[View More Details](#)

← Past Export Price Trends

Historical Export Price Analysis



Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut. Dolore exercitation nulla tempor consequat .

Export Details By Country



Labore sunt veniam amet est. Minim nisi dolor eu ad.

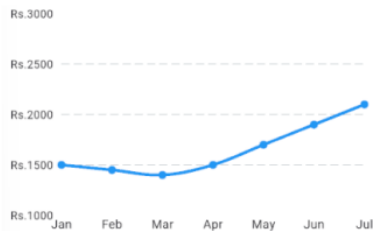
View →

Filter Data

Year	From	To
	Jan	Jul

Global Black Pepper Price Trends

Jan - Jul



Peak Price

Rs. 2100/kg

Lowest Price

Rs. 1400/kg

Average

Rs. 1650/kg

Key Insights

Prices are trending upward

Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut.

Peak vs Average

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Recommendation

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← Past Export Price Trends

Historical Export Price Analysis



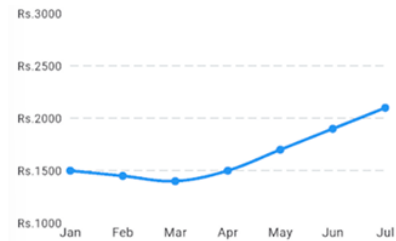
Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut. Dolore exercitation nulla tempor consequat .

Filter Data

Year	From	To
	Jan	Jul

Global Black Pepper Price Trends

Jan - Jul



Peak Price

Rs. 2100/kg

Lowest Price

Rs. 1400/kg

Average

Rs. 1650/kg

Key Insights

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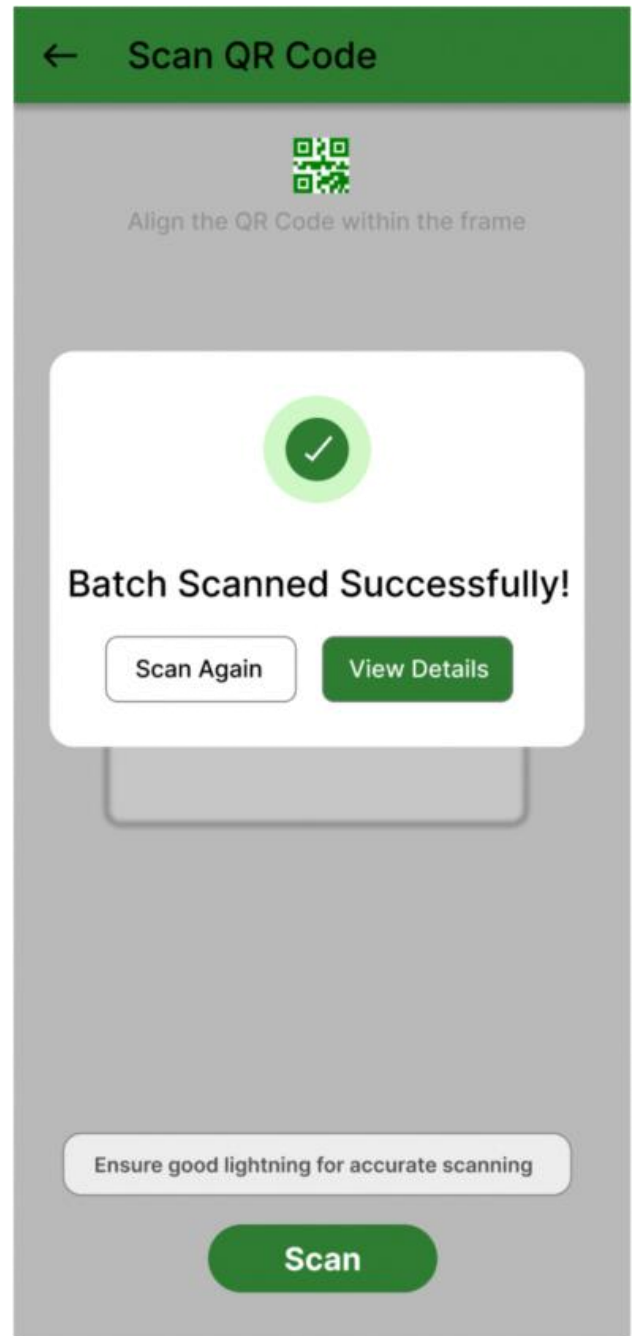
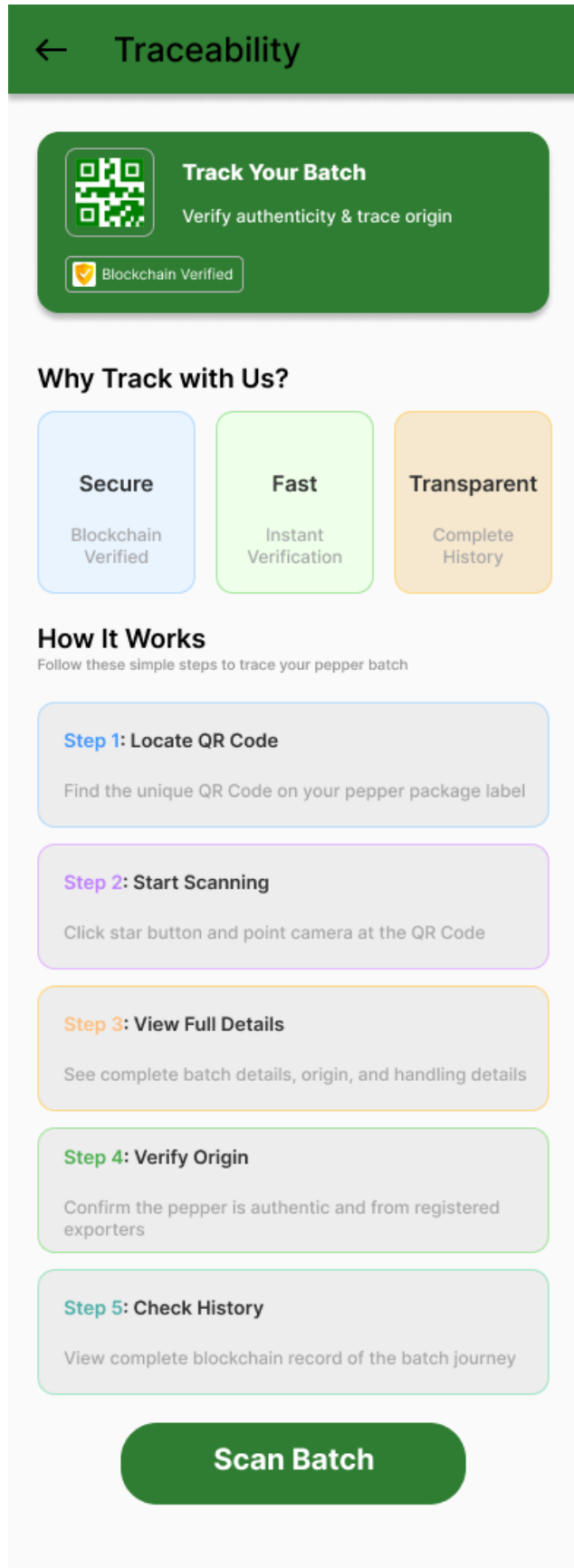
Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut.

Momentum & Stability

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Recommendation

Labore sunt veniam amet est. Minim nisi dolor eu ad incididunt cillum elit ex ut.



< Quality Grading



24

Total Reports



8

Premium Grades

Grading Actions



New Quality Check

Start Grading →



Past Reports

View History →



How It Works

Learn Process →



Quality Tips

Improve Grade →

Resources & Support



IoT Device Setup

Connect your ESP32 bulk density device >



Image Capture Guide

Learn how to capture quality images >



Certification

Upload and verify your certificates >



Dashboard



My Farm



Market



Profile

< Batch Details

1

2

3

Batch Information

Enter your pepper batch details

Pepper Information

Pepper Type

Black Pepper

Pepper Variety

Ceylon Pepper

Harvest & Processing

Harvest Date

Select Harvest Date

Drying Method

Sun Dried

Certificates & Compliance

GAP



Certificate of Origin (COO)



Quarantine Certificate



Quality Certificate



ISO 22000



HACCP (SLS 1266)



FSSC 22000



Organic Certification



Fair Trade



Quantity

Batch Weight (Optional)

Kilograms

Grams

Enter weight in kg and/or grams

Continue to Bulk Density →

<

Bulk Density

✓

2

3

Density Measurement

Connected

Read density from IoT device

i

Measurement Instructions

1. Ensure IoT device is connected

2. Place pepper sample in the device

3. Click "Read from Device" button

4. Wait for the reading to complete

?

View detailed Instructions

📶

Bulk Density Reading

Not ready yet

Click "Read from Device" to measure

Waiting for reading

Read from Device

Continue to Image Upload →

<

Image Upload

✓

✓

3

Capture Pepper Images

Take 9 images from different angles and layers

i

Image Capture Instructions

1. Place pepper on clean white A4 paper

2. Ensure good natural lighting

3. Capture 9 images (3 angles*3 layers)

4. Keep camera 20-30 cm above samples

?

View detailed Instructions

↓

Bottom Sample

📷

Full View

📷

Half View

📷

Close-up

—

Middle Sample

📷

Full View

📷

Half View

📷

Close-up

↑

Top Sample

📷

Full View

📷

Half View

📷

Close-up

Continue →

23 | Page

Quality Report

Premium

90/100

Overall Quality Score

Batch Information

Pepper Type	Black Pepper
Pepper Variety	Ceylon Pepper
Drying Method	Sun Dried
Batch Weight	25 kg
Bulk Density	540 g/L
Certificates	GAP, Quality Certificate

Quality Breakdown

Size / Pinheads92

Color Uniformity88

Surface Defects94

Extraneous Matter98

Adulteration100

Uniformity90

Improvement Suggestions

Ensure uniform drying to improve color score

Remove broken berries before packing

Next Steps

Check Market Price
Get current rates

Start New Batch
Grade another batch

Download Report (PDF)

How is Quality Calculated?

< Quality Tips

Explore Quality Factors

Variety & Piperine

Learn more →

Color Uniformity

Learn more →

Size & Shape Consistency

Learn more →

Mold Prevention

Learn more →

Drying Process

Learn more →

Storage Tips

Learn more →

Quick Tips

✓ Labore sunt veniam amet est.

✓ Labore sunt veniam amet est.

✓ Labore sunt veniam amet est.

Figma Link

<https://www.figma.com/design/teuLPeIljJqqiK0m8e135d/Research-Project?node-id=0-1&t=XdTQQELgs7rD2wSQ-1>

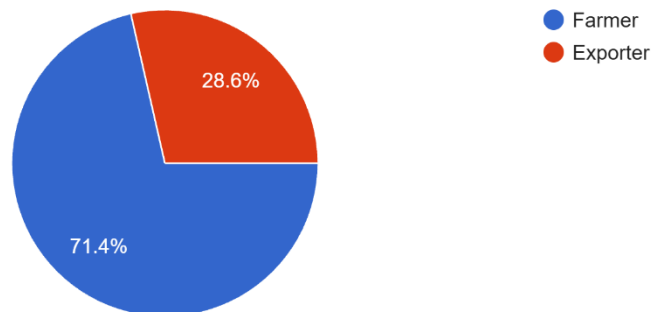
User Feedback

Google Form link - <https://forms.gle/jq5yWPP1fRukD6to9>

These are the responses we received,

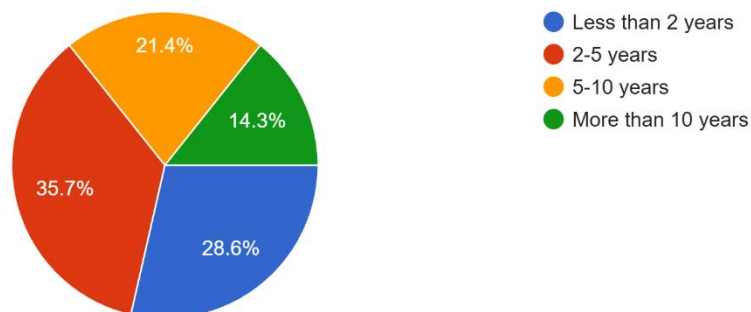
Select role

14 responses



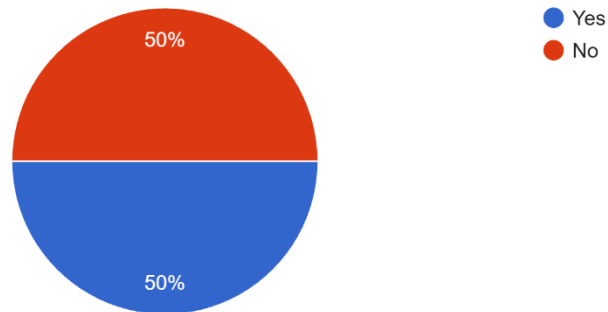
Years of experience in agriculture

14 responses



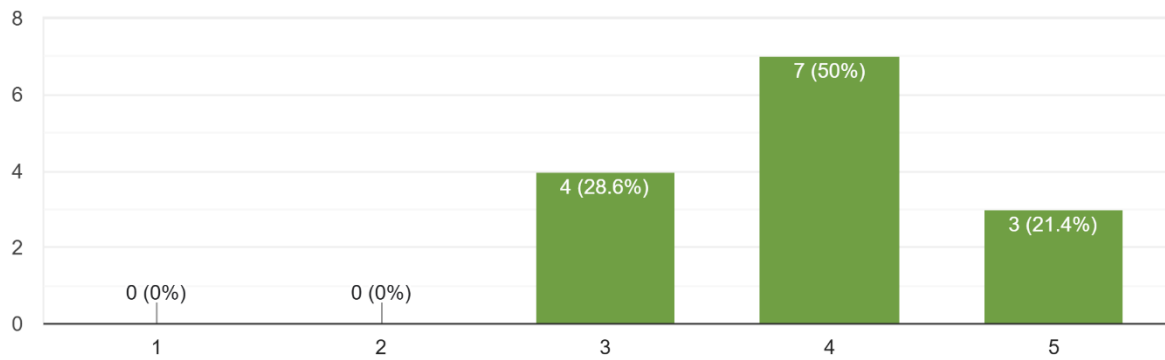
Have you used mobile apps for agriculture before?

14 responses



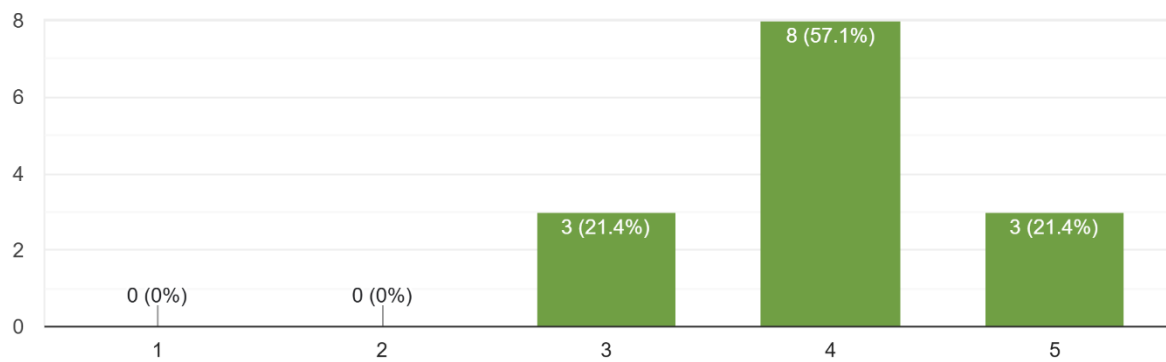
How easy was it to use the application? (Very difficult to very easy)

14 responses



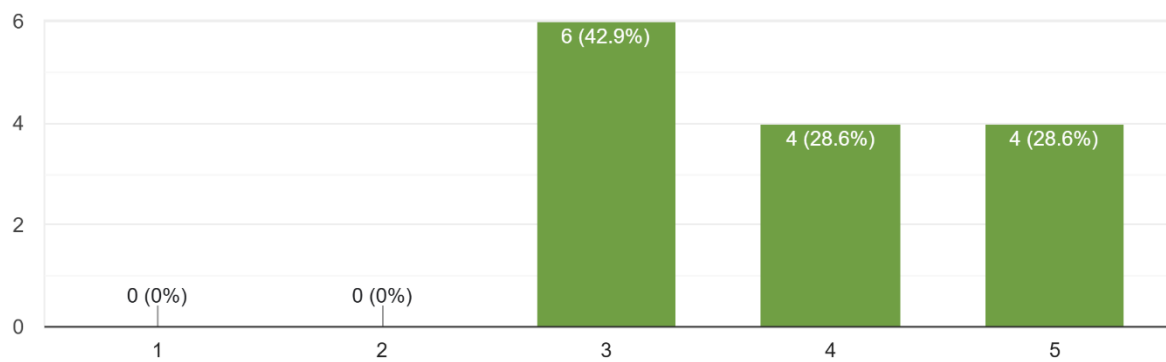
How clear was the overall layout and navigation of the app?

14 responses



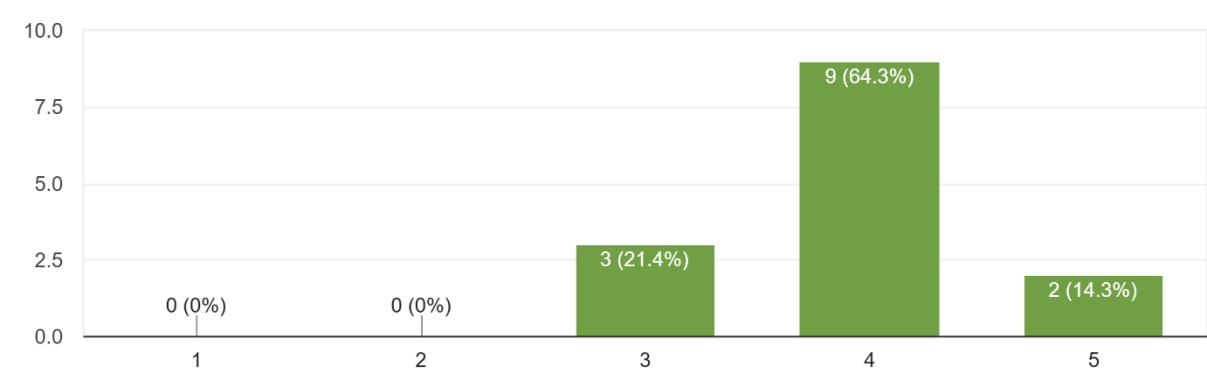
How satisfied are you with the prototype overall?

14 responses



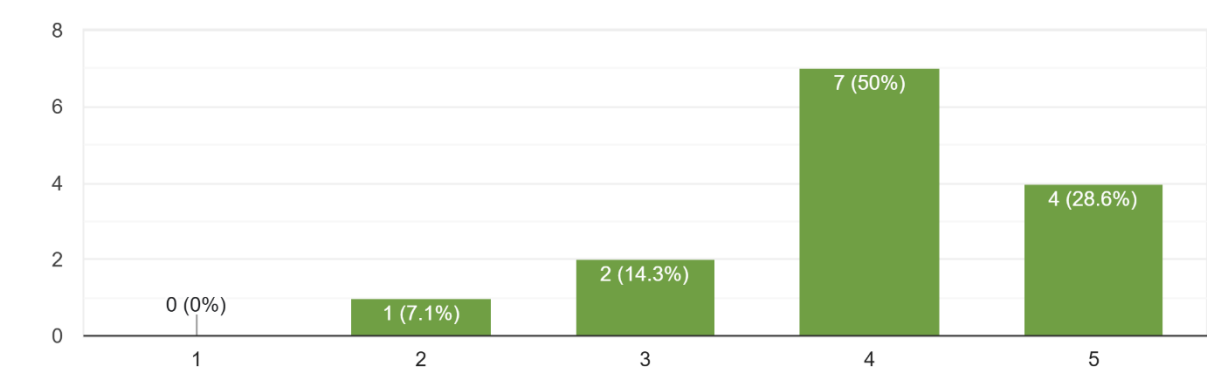
The app interface is visually clear and well-organized (Strongly disagree to strongly agree)

14 responses



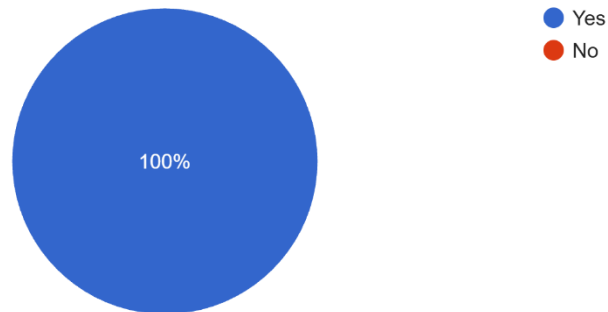
Text size, icons, and colors are easy to understand(Strongly disagree to strongly agree)

14 responses



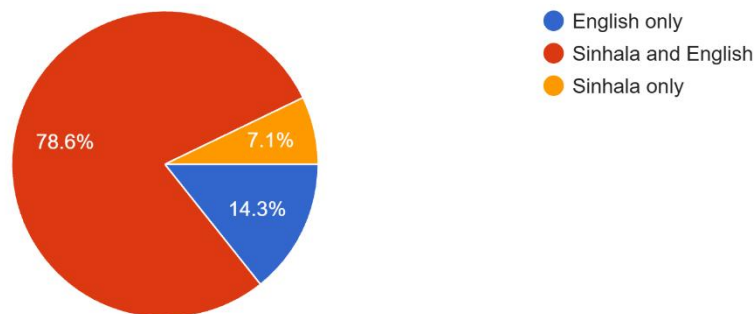
Navigation between screens (dashboard, prediction, history, insights) was smooth

14 responses



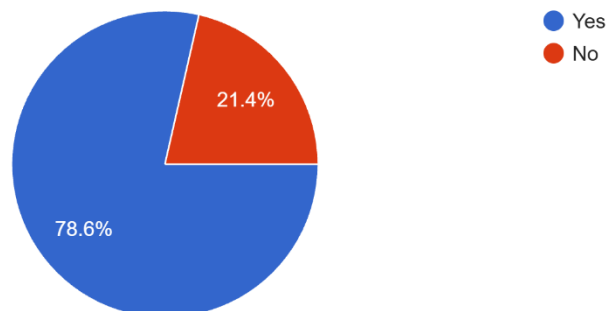
Which language support would you prefer?

14 responses



Would Sinhala language support improve usability for you?

14 responses



Improvements & Suggestions

What features would you like to see added to this application?

5 responses

weekly export price

like to see in sinhala language

I would like to have export prices destination wise

Since we have no knowledge about emails, can u add mobile phone option for login and signup

Better to have a market place to sell your products too

What was the biggest challenge you faced while using the prototype?

1 response

navigation issues

Improvements based on user feedback

- **Sinhala Language Support**
Add full Sinhala language option to make the application more accessible for farmers who are not comfortable with English.
- **To have mobile number instead of email for account creation**
Enable users to register and log in using their mobile phone number instead of email, as many farmers are not familiar with using email. This will simplify the registration process and reduce barriers to entry.
- **Error Messages & Feedback**
Show simple, clear error messages and success messages in a user-friendly way.
- **Destination wise export price**
- **Marketplace to sell spice items, equipment, and fertilizers**